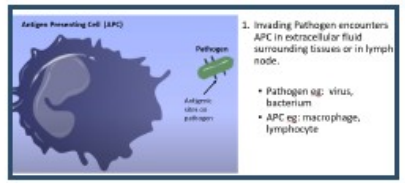
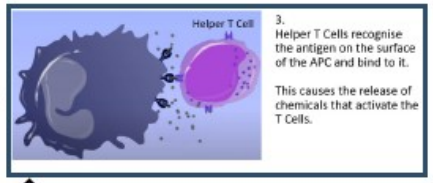
Flow Diagram— Specific Immunity



1. Invading Pathogen encounters APC in extracellular fluid surrounding tissues or in lymph node.
   - Pathogen eg: virus, bacterium
   - APC eg: macrophage, lymphocyte



2.
The APC engulfs the pathogen ...
... and processes it...
... and presents the antigen on its surface



3.
Helper T Cells recognise the antigen on the surface of the APC and bind to it.

This causes the release of chemicals that activate the T Cells.

Steps 1-4 show the Activation Phase of specific immunity. This is where T-cells are activated and proliferate. Then, Humoral and Cell Mediated immunity take place to deal with the pathogen both outside and inside of body cells.



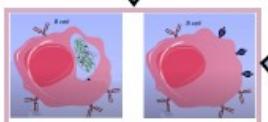
4.
T-cells divide and proliferate into different types, producing many that have the specific receptor for the antigen.

T-cell types include:
- Helper T-cells
- Cytotoxic T-cells
- Suppressor T-cells
- Killer T-cells
- Memory T-cells

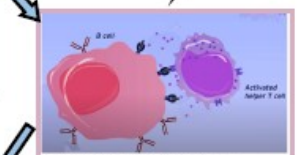
Humoral Immunity - deals with pathogens in extracellular fluids via production of antibody . "Antibody Mediated Immunity"



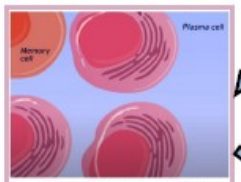
A B-cell with the same receptor as for the original antigen binds to the pathogen.
- Both B and T Lymphocytes express many receptors on their surface, that match many pathogens.



The B-cell engulfs the pathogen, processes it and presents the antigen on its surface.



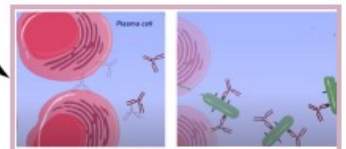
A helper T-cell once again binds, and releases chemicals that stimulate the B-cell



The stimulated B-cell then divides, and proliferates into plasma cells, and B-memory cells.



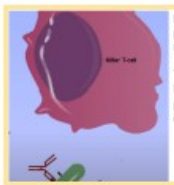
B-memory cells remain, so that antibody can be produced much more quickly next time the specific pathogen is encountered.



The plasma cells now produce large amounts of antibody specific to the original pathogen.

This then circulates in body fluids, and neutralises and removes pathogens.

Several chemicals are also produced (including complement) that assist in destroying pathogens.



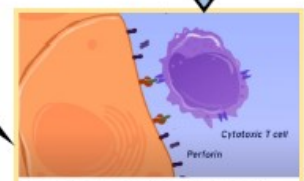
Killer T-cells phagocytose pathogens that have been neutralised by antibody.

They also produce proteins that directly destroy some pathogens, or cells that contain pathogens

Cell Mediated Immunity - deals with pathogens inside cells by identifying infected cells and destroying them

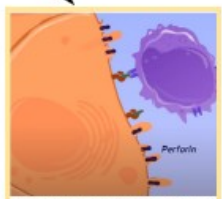


The pathogen invades a body cell. The body cell processes it and presents the antigen on its cell surface.

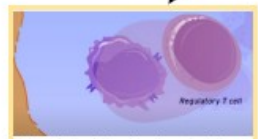


Cytotoxic T-cells activated during the Activation Phase bind to the antigen on the cell surface, and produce a chemical called perforin

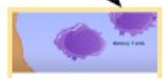
Note: some Cytotoxic T-cells are especially adapted to identify and destroy some cancer cells.



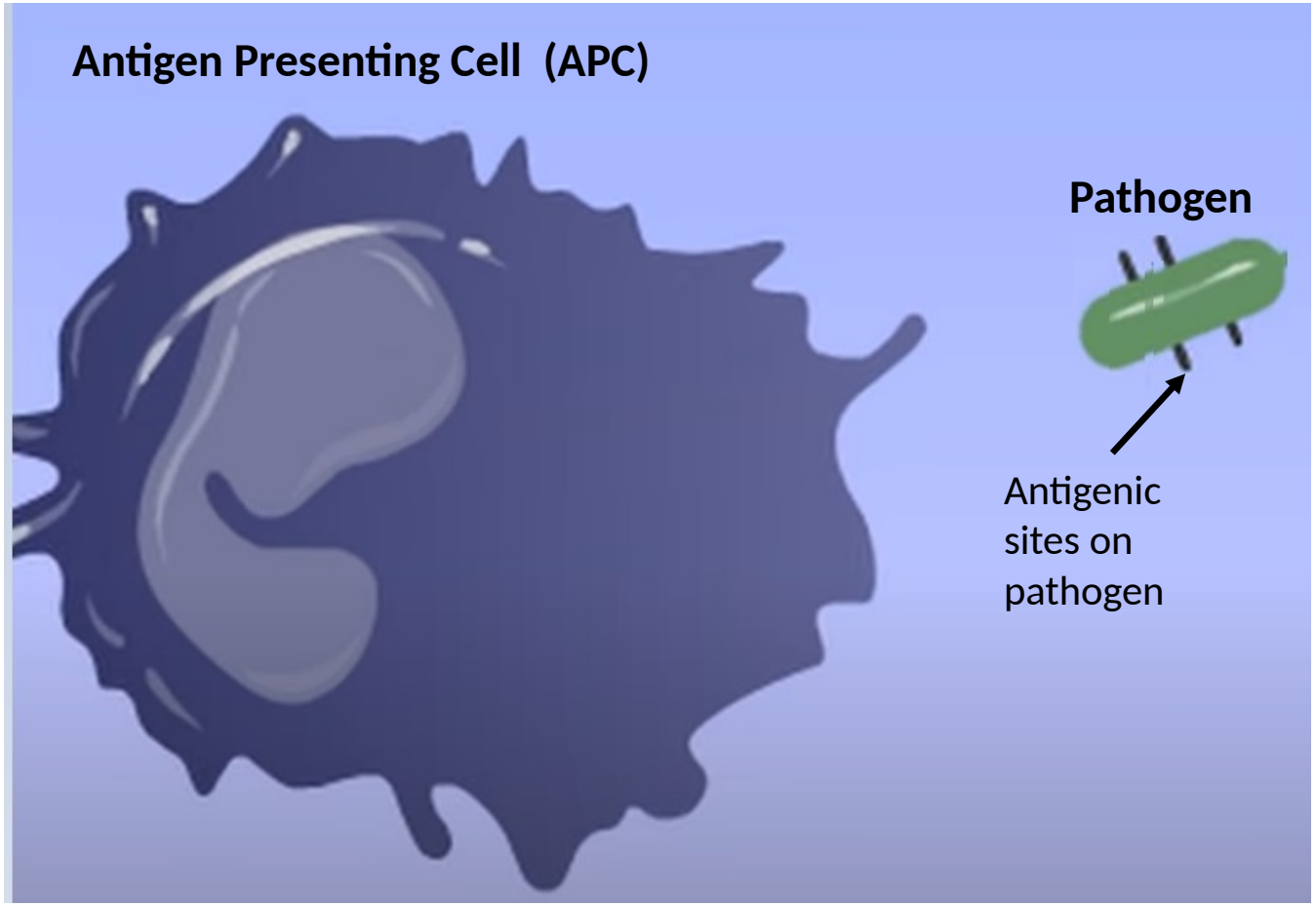
Perforin causes the infected cell to rupture and die, also killing the pathogens inside.



As the infection is brought under control, the activated Cytotoxic T-Cells are turned off by Regulatory (Suppressor) T-Cells.



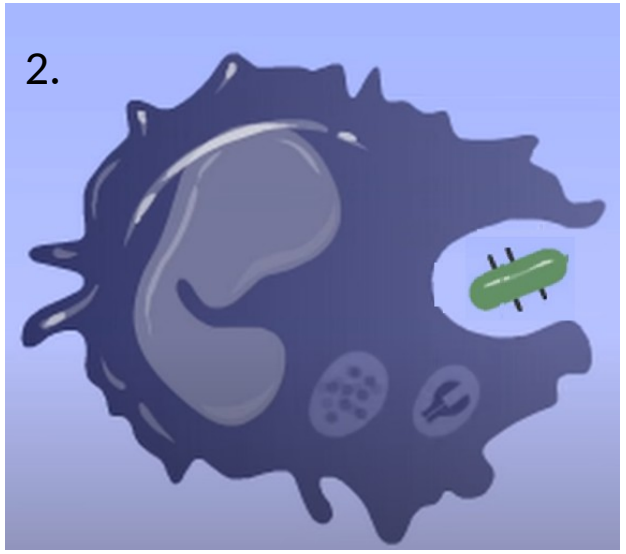
Memory T-cells stay in circulation so that the response will be faster next time the pathogen is encountered.



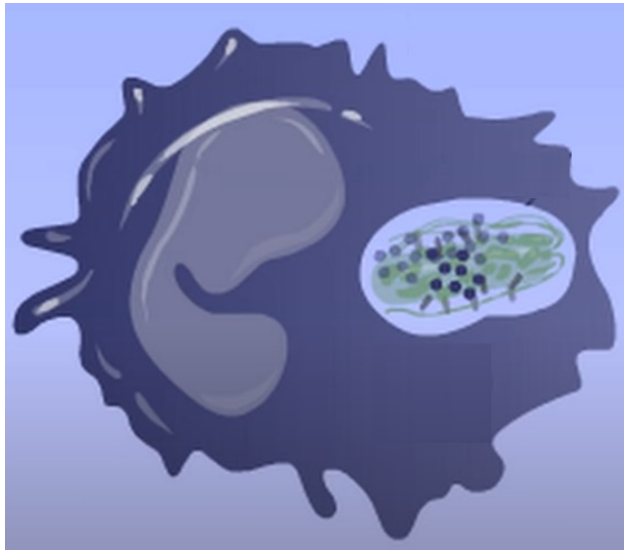
1. Invading Pathogen encounters APC in extracellular fluid surrounding tissues or in lymph node.

- Pathogen eg: virus, bacterium
- APC eg: macrophage, lymphocyte

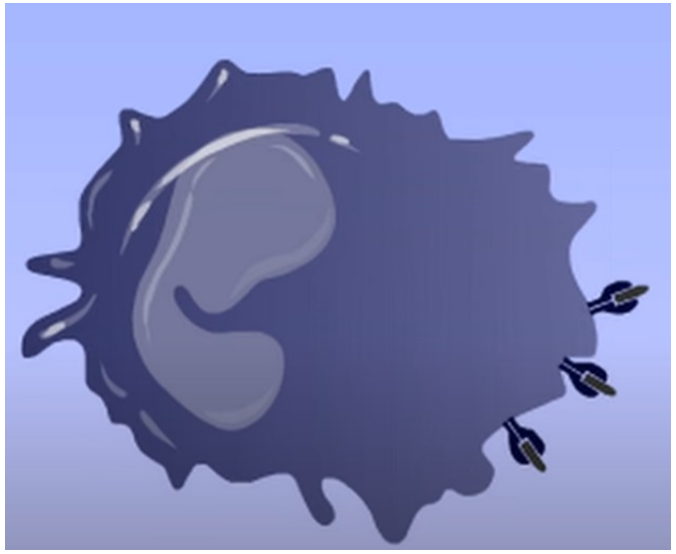
2.



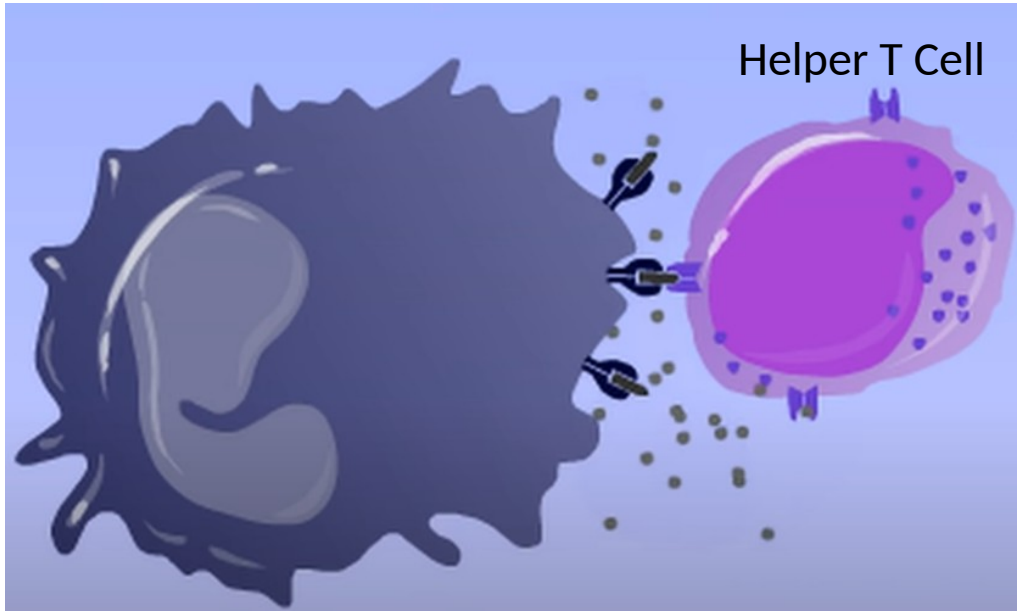
The APC engulfs the pathogen …



… and processes it…



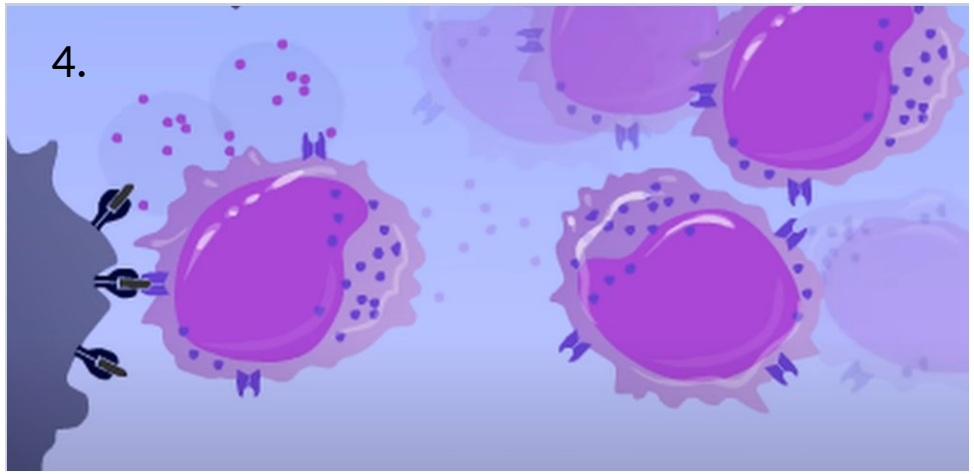
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3.
Helper T Cells recognise the antigen on the surface of the APC and bind to it.

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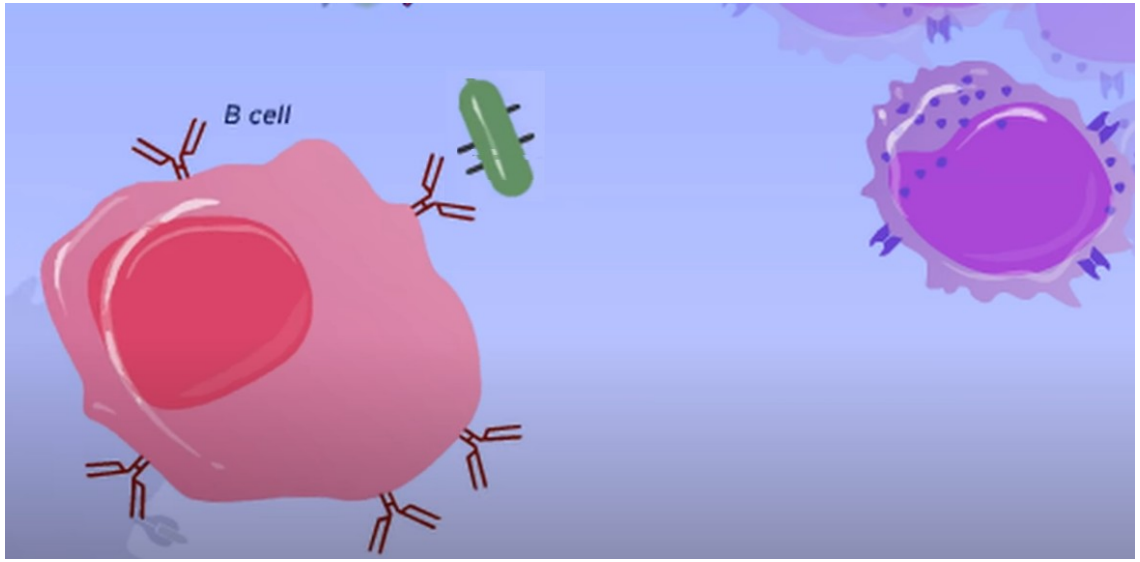
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T-cells divide and proliferate into different types, producing many that have the specific receptor for the antigen.

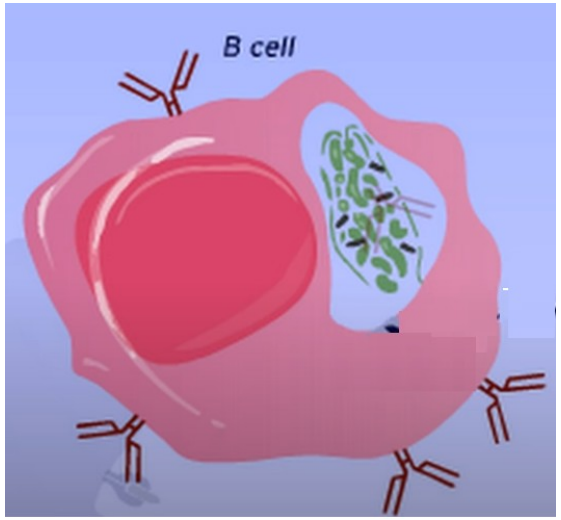
T-cell types include:

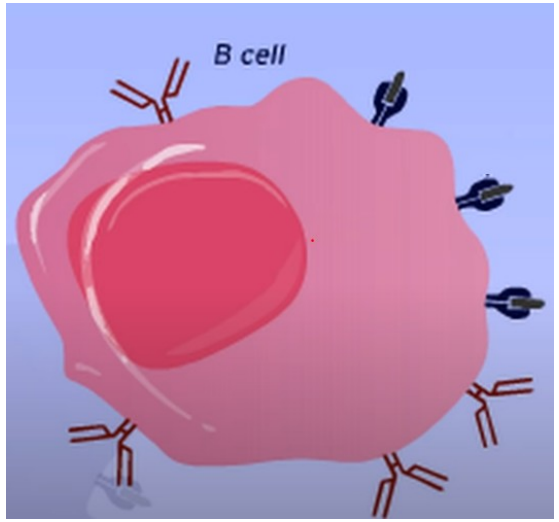
- Helper T-cells
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- Memory T-cells



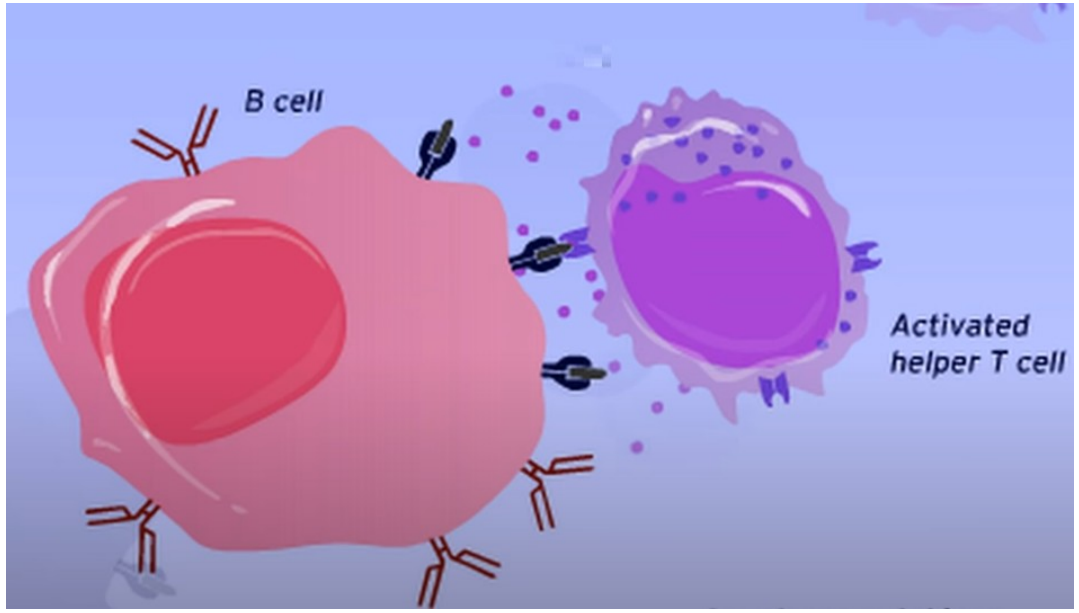
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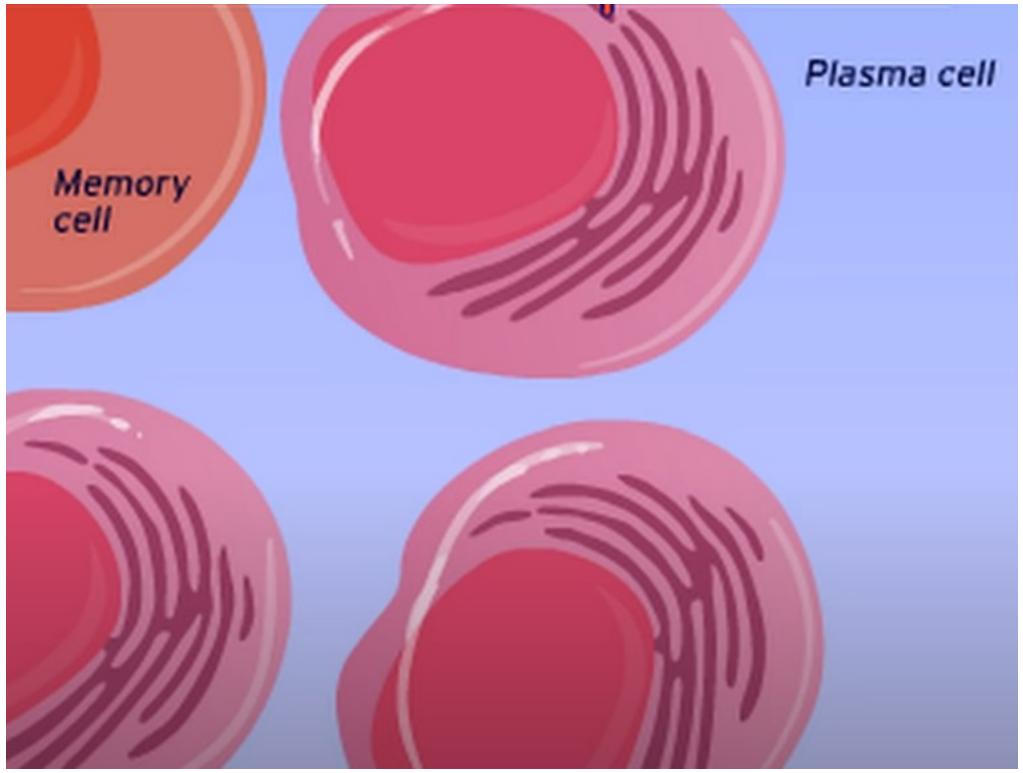




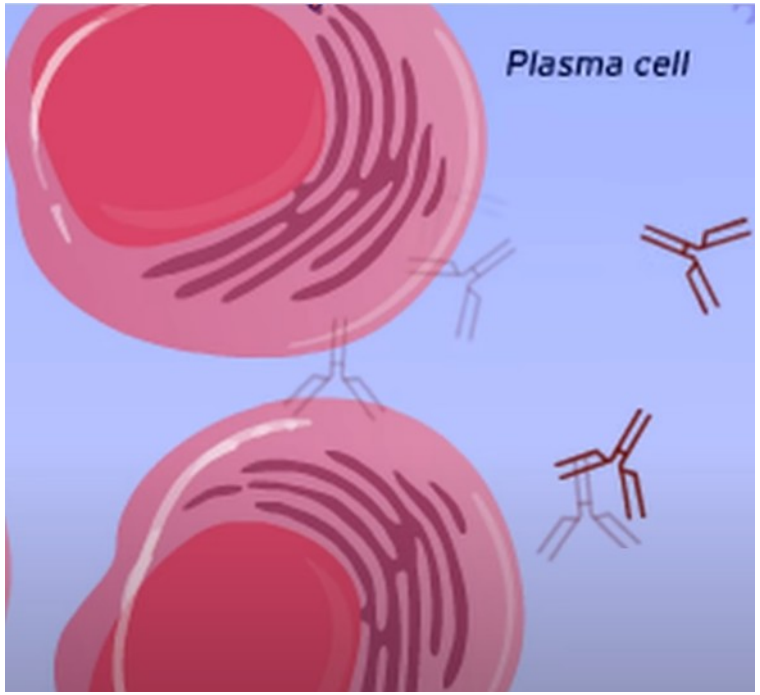
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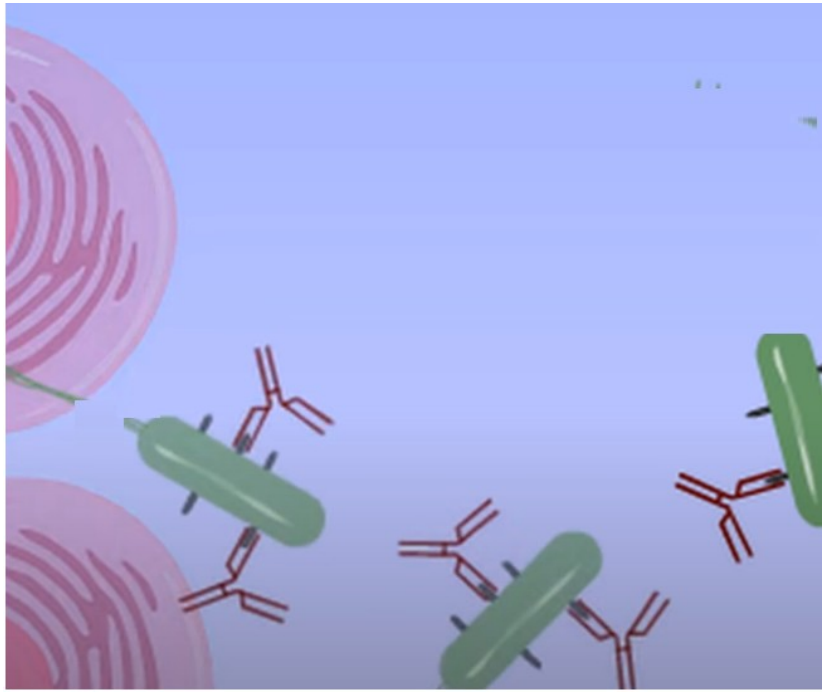


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4. The stimulated B-cell then divides, and proliferates into plasma cells, and B-memory cells.

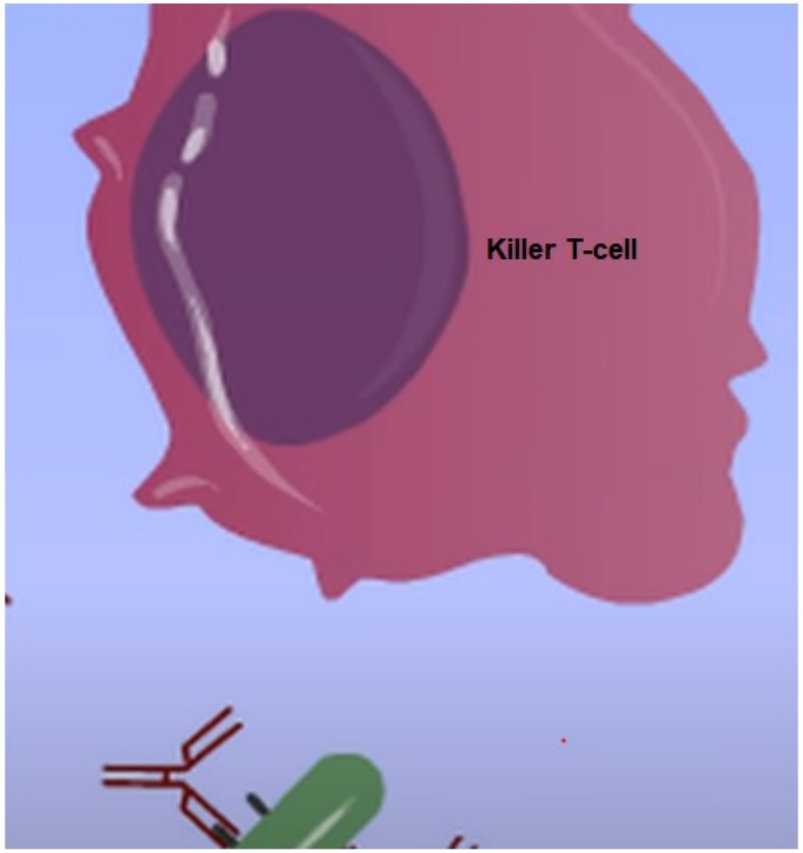




The plasma cells now produce large amounts of antibody specific for the original pathogen.

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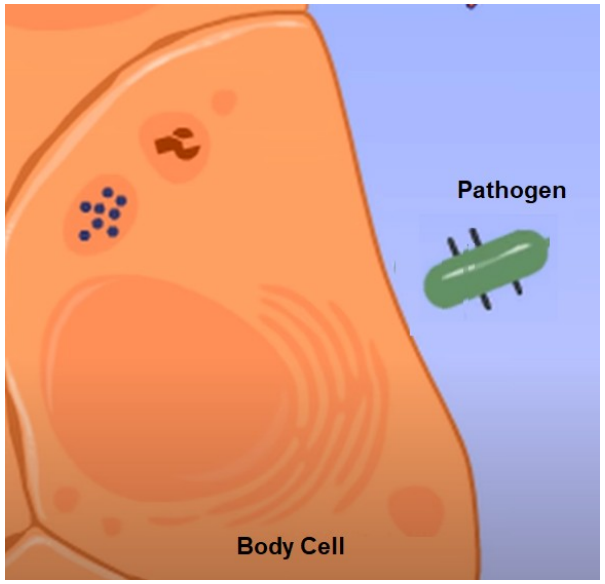


Killer T-cells phagocytose pathogens that have been neutralised by antibody.

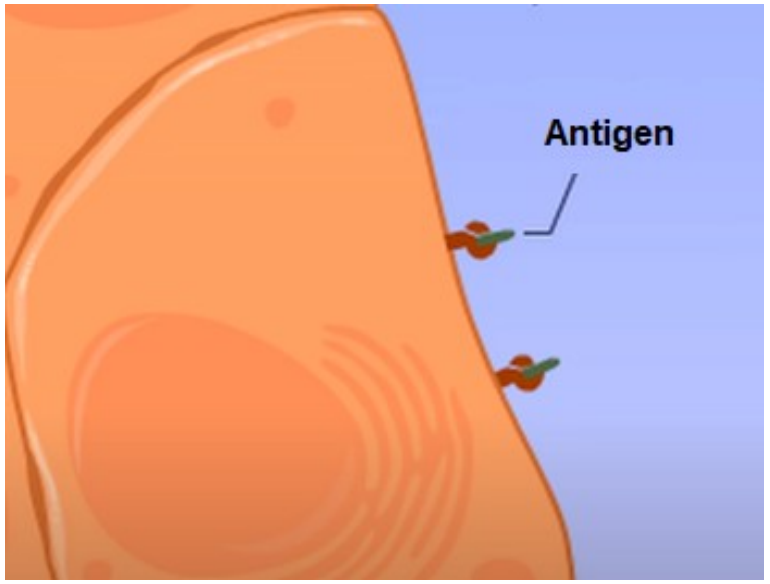
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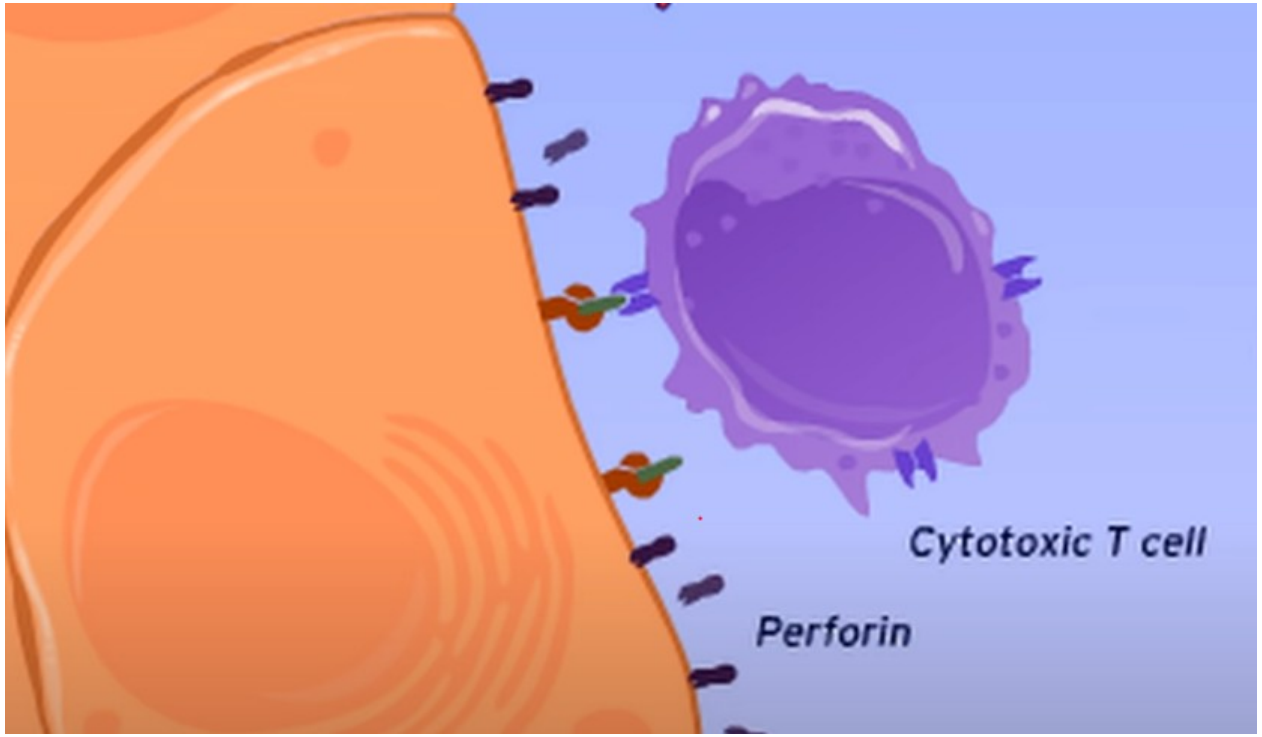
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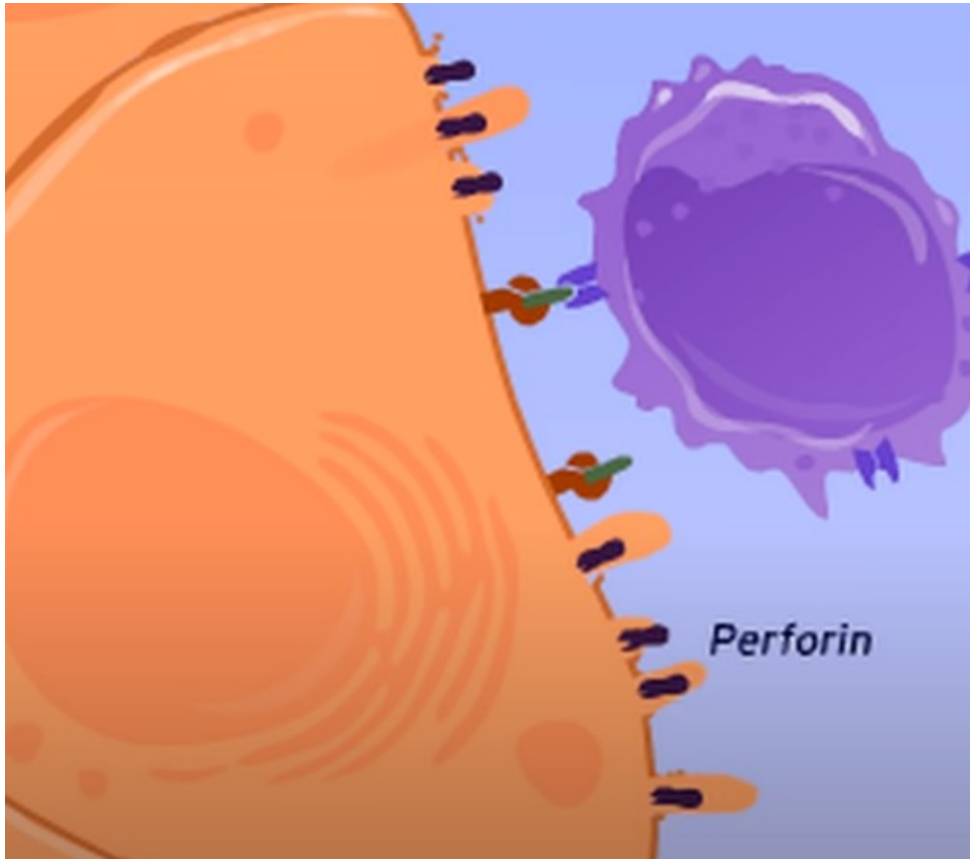




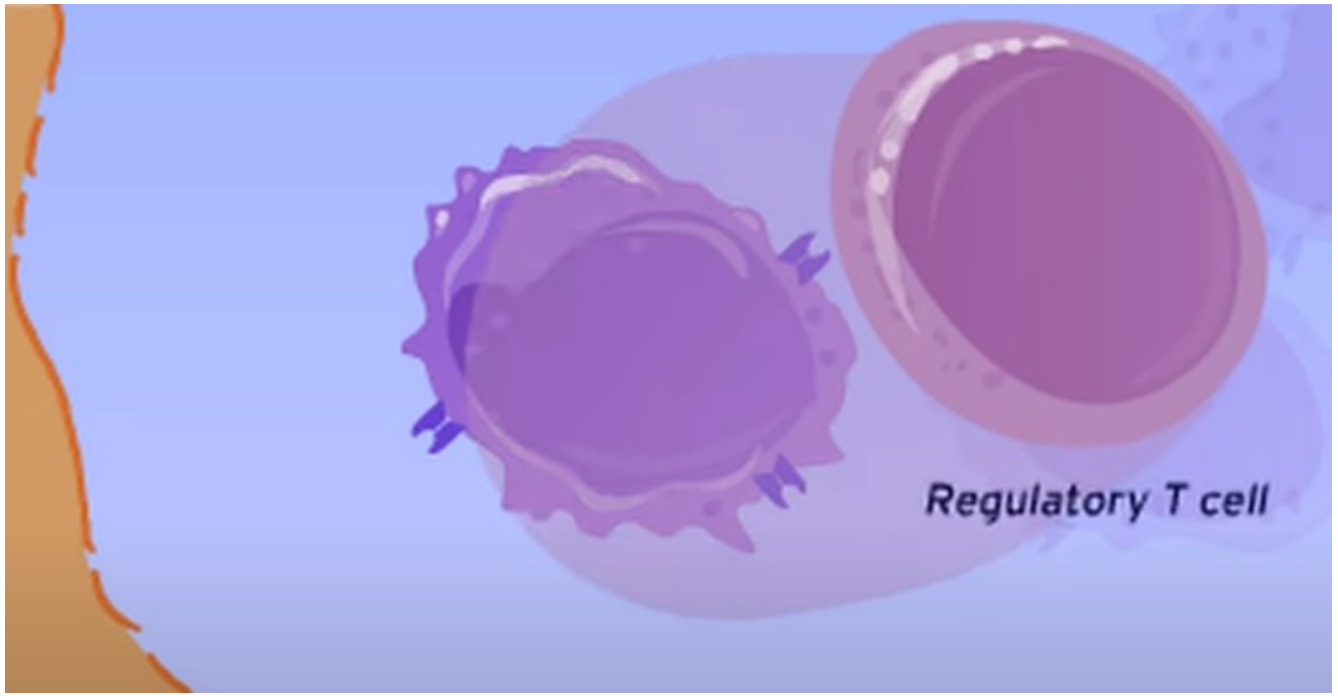
The pathogen invades a body cell. The body cell processes it and presents the antigen on its cell surface.



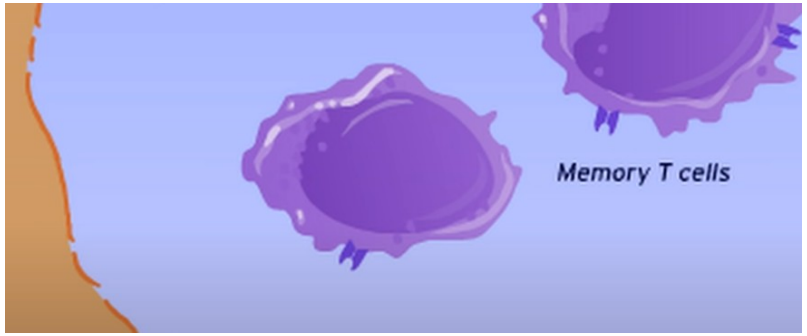
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Perforin causes the infected cell to rupture and die, also killing the pathogens inside.



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